Lab 9 - Support Structures for Overhangs

Course: Innovating Product Development with 3D Printing and Additive Manufacturing (TGS-2022017597) Topic 02: Process Related Issues for AM TSC objective: A2, A3 - determine when overhang geometry requires supports and generate them

Goal

Use supplied twist, wedge and sideways-hole models to identify unsupported islands, compare orientation choices, then generate only the supports the geometry needs.

What you'll build

A support decision sheet plus a printed Twisted Solid whose critical overhangs succeed with controlled support use.

Tools: Ultimaker Cura, Twisted-Solid.stl, wedge.stl, cube20-sideways-hole.stl, FDM printer

Steps

1. Open Twisted-Solid.stl in Cura and slice it with supports off.

```
File > Open > Twisted-Solid.stl > Slice
```

2. In Preview, find the layers that start in mid-air - these overhangs would droop or fail.

```
Preview > layer slider - spot unsupported islands
```

3. Enable support generation.

```
Print Settings > Support > Generate Support ✔
```

4. Set placement to Touching Buildplate first; compare with Everywhere and note the difference in Preview.

```
Support Placement - Touching Buildplate vs Everywhere
```

5. Keep the support overhang angle at 45-50° so only genuinely steep faces get support.

```
Support Overhang Angle - 50 degrees
```

6. Slice again and compare time and material with the unsupported version, then print.

```
Slice > Save to Disk > print
```

7. Snap the supports off with pliers, working from the edges inward, and inspect the supported surfaces.

```
Needle-nose pliers > remove supports
```

8. Repeat the analysis with wedge.stl and cube20-sideways-hole.stl; for each, compare adding support with rotating the model to make the geometry self-supporting.

```
Open both models > Rotate > Slice > Preview > record decision
```

Evidence to capture

- Three-model support decision table: risk, best orientation, support placement and reason.
- Before/after Cura Preview screenshots for Twisted-Solid.stl with supports off and on.
- Photo of the printed benchmark after support removal, with scarring annotated.

Test it

Your decision table correctly identifies unsupported islands and distinguishes when support is necessary from when reorientation is the better fix; the Twisted Solid print completes with removable support and only minor underside scarring.

Files in this folder

- twist.stl - Core sample model used by the lab procedure.
- Twisted-Solid.stl - 49 mm twisted benchmark with steep faces and changing overhangs.
- wedge.stl - Simple 60 mm overhang-angle benchmark for reading progressive underside quality.
- cube20-sideways-hole.stl - Sideways-hole benchmark for comparing support, bridging and reorientation.